

$$10. \quad v = h(p(u))$$

$$\frac{dv}{du} = h'(p(u)) \cdot p'(u)$$

$$\frac{dv}{du} = h'(p(u)) \cdot \left(\frac{3}{2} u^{\frac{1}{2}} + 9u^2 - 5 \right)$$

$$\left. \frac{dv}{du} \right|_{u=1} = h'(p(1)) \cdot \left(\frac{3}{2} + 9 - 5 \right)$$

$$\left. \frac{dv}{du} \right|_{u=1} = h'(-1) \left(\frac{11}{2} \right)$$

$$\left. \frac{dv}{du} \right|_{u=1} = \frac{55}{2}$$

$$11. \quad y = ax^3 - 4a^2x - \frac{x}{2a-3b}$$

$$a) \quad \frac{dy}{dx} = 3ax^2 - 4a^2 - \frac{1}{2a-3b}$$

$$b) \quad \frac{dy}{da} = x^3 - 8xa + \frac{2x}{(2a-3b)^2}$$

(8)

$$12. \quad f(x) = x^3 - 9x$$

$$f'(x) = 3x^2 - 9$$

$$y - 16 = (3x^2 - 9)(x - 0)$$

$$x^3 - 9x - 16 = 3x^3 - 9x$$

$$2x^3 = -16$$

$$x^3 = -8$$

$$x = -2 \Rightarrow f'(-2) = 3$$

$$\therefore y - 16 = 3x$$

$$\text{let } y = 0 \Rightarrow x = -\frac{16}{3}$$

$$\text{Area} = \frac{1}{2} b \cdot h$$

$$\text{Area} = \frac{1}{2} \left(\frac{16}{3} \right) (16)$$

base is positive quantity (ie: $+\frac{16}{3}$)

$$\text{Area} = \frac{128}{3} \text{ units}^2$$

